D122 | Information needs a later choice

PRIVATE FICTIONAL two-state scale/stop case. All money lakh INR at one common present-value date. No actual research purchase, pilot, scale-up, financing, order, contact, disclosure or spend.

Attempt first	Construct the joint tree, Bayes posteriors and optimal actions before reading references. A risk-neutral expected monetary-value criterion is stipulated, not a guarantee, risk-adjusted utility, financing feasibility or actual approval.
No double capex	Scale net terminal values +100/-40 already include investment and operating cash flows. Stop is0. Best no-test scale EV16 retains downside-40. Do not subtract scale capex a second time.
Signal before decision	Prior success.4, positive rates.8 if success/.2 if failure. Joint .32/.12/.08/.48. Positive probability.44, posterior8/11; negative probability.56, posterior1/7. Sensitivity80% is not posterior72.727273%.
Value is action-dependent	After positive scale EV61.818182; after negative scale EV-20 so stop. Gross optimal signal payoff27.2; EVSI11.2. Perfect-info gross40; EVPI24. Signal need not be perfect to equal perfect information in decision value.
Cost and delay separately	Acquisition5 plus present-valued delay/opportunity2 each paid once on every test path. Net test-policy20.2; gain4.2 over no-test16. Conditional cost ceiling9.2. Acquisition10 makes policy15.2, inferior despite positive gross EVSI.
Too late to help this choice	Once scale is irrevocable and no action/payoff can change, information-only decision value is0; paying5+2 reduces committed scale EV16 to9. Other later uses require defined new decisions, not invented benefits.
Scope and evidence	Static prose is Case-v1, revise after edits. Signals do not causally alter product/payoff in this model. Selected source text only, not all diagrams/full AV. Actual attempts and independent learning unobserved. No overall10 or account clearance.

Inputs | Fixed state, optional information

Synthetic probabilities and NET terminal present values. Stop payoff fixed0. Success payoff>=0, failure<=0, at least one strict. Costs nonnegative and paid once when test acquired.

Prior success probability	0.40
Scale success net payoff, lakh INR	100.00
Scale failure net payoff, lakh INR	-40.00
Positive signal probability given success	0.80
Positive signal probability given failure	0.20
Acquisition cost, lakh INR	5.00
Separate PV delay/opportunity cost, lakh INR	2.00

Decision model inputs valid? (1=yes)	1.000000
Test costs valid? (1=yes)	1.000000

Valid numerical bounds do not verify empirical priors, likelihoods, project cash flows or risk capacity. Costs are separate from terminal values only by explicit assumption. If a pilot changes state, performance, cash flows or available actions, rebuild the causal model.

Joint tree | Multiply then condition

Joint = prior state probability times signal likelihood. Each column is one mutually exclusive signal. Impossible signal has undefined posterior, not zero success probability. No rounded posterior feeds the model.

Joint probability	Positive signal	Negative signal
Success joint	32.0000%	8.0000%
Failure joint	12.0000%	48.0000%
Marginal signal probability	44.0000%	56.0000%
Posterior success, if signal possible	72.7273%	14.2857%
All joint probabilities sum	1.000000	

Baseline positive .32+.12=.44, posterior .32/.44=8/11. Negative .08+.48=.56, posterior .08/.56=1/7. Signal rates are assumed conditional probabilities, not measured accuracy. Zero-probability branches carry zero weighted value and no chosen action.

Foldback | Preserve scale and stop

Conditional EV before acquisition/delay costs. Fixed stop0. Zero-probability signal is UNREACHED; no division by zero. Weighted optimal branch value uses joints directly for stability.

Measure	Positive signal	Negative signal
Conditional scale EV, lakh INR	61.818182	-20.000000
Optimal action before test costs	SCALE	STOP
Weighted optimal payoff, lakh INR	27.200000	0.000000
Prior scale EV, lakh INR	16.000000	
Best no-test EV, lakh INR	16.000000	
No-test action	SCALE	
Scale break-even success probability	28.5714%	
Gross optimal with-signal EV	27.200000	

Baseline no-test16; positive SCALE61.818182/negative STOP0; weighted27.2. Costs are sunk at signal time but matter when deciding whether to acquire. Tie means either scale or stop yields the same EV under the stated criterion, not identical downside.

Value | Gross information versus net gain

All amounts lakh INR at the same valuation date. Delay is separately given present value, not a new discount rate. Gross EVSI/EVPI independent of acquisition cost in this fixed model.

Gross expected value of sample information	11.200000
Gross optimal payoff with perfect information	40.000000
Expected value of perfect information	24.000000
Acquisition plus delay, paid once	7.000000
Net test-policy EV	20.200000
Net gain versus best no-test	4.200000
Acquisition cost ceiling with fixed delay	9.200000
Acquire under EV criterion?	PREFER TEST
Gross bounds 0<=EVSI<=EVPI? (1=yes)	1.000000

A negative cost ceiling means even a free acquisition is inferior once stated delay is paid. Equality is a net-EV tie, not approval. Positive EVSI alone does not justify purchase. EVSI can equal EVPI for perfect or decision-equivalent sample information; zero EVSI is possible.

Too late | Scale already irrevocable

A separate specified decision: scale was committed, both terminal payoffs and all later actions fixed. Information only changes beliefs. It cannot be credited with an unspecified future use.

Committed scale EV before test	16.000000
Committed scale EV with free information	16.000000
Information decision value after commitment	0.000000

Committed EV after paid test and delay	9.000000
Change caused by acquisition costs	-7.000000

If prior scale EV becomes negative in a changed case, the committed path keeps that negative value: it cannot silently choose stop. Compare this path to its own no-information committed baseline, not the still-reversible best no-test policy. Baseline16 ->9 after5+2.

Audit | Algebra is not an approval

Independent gross and cost checks. Bad costs do not erase valid gross information arithmetic. Broken critical references hold affected summaries. No empirical signal calibration or real authority is verified.

Gross joint/foldback identities valid? (1=yes)	1.000000
Guarded EVSI, lakh INR	11.200000

Net cost identities valid? (1=yes)	1.000000
Guarded net gain, lakh INR	4.200000

Tests cover defined identities, not every possible formula corruption. Valid arithmetic cannot establish real payoffs, independent signal rates, permission, risk acceptance or a worthwhile pilot. Separate source evidence and authority would be needed for actual activity.

Evidence | Scope and strongest objections

Selected Dartmouth teaching text rechecked13 November2026. Not full visual slide/diagram or full video coverage. No real technical validation, participant, research purchase or project observation.

Teaching source	https://cushman.host.dartmouth.edu/courses/engs41/Value-of-info.pdf - joint probabilities, Bayes reversal, foldback, EVSI before sampling cost and expected net gain. Source uses rounded oil example; those numbers are not copied into this synthetic two-state case.
Bound correction	Source says sample information always less valuable than perfect information. The correct bound permits equality under the same model:0<=EVSI<=EVPI. Perfect signal or identical optimal decisions can make equality; no free guarantee of a successful project follows.
Purchase correction	Source sentence says EVSI>0 supports purchase. That is not sufficient with positive acquisition/delay costs; compare EVSI to all separately modeled costs. This workbook keeps gross and net values distinct.
Pure information assumption	If a pilot changes success probability, product performance, scale cost, salvage or later actions, evaluate those causal effects and costs explicitly. Do not label every improvement EVSI or both subtract a cost already included in terminal values and subtract it again.
Day0 hold gate	Fictional technical owner checks state definition/signal rates; cash owner checks common valuation date and complete costs; governance owner checks rights/risk/permission. No real owner accepted these roles. Missing evidence or authority holds actual activity.
Day30 review	Record actual objection, model revision and new changed-context transfer only if performed. Check irreversibility and ability to stop before buying information. No invented reviewer acceptance, pilot outcome, learner mastery or account test. Static prose remains Case-v1.

Actual work | No observed pilot or learner record

Record only actual work performed, not synthetic outcomes as realized results. Blank fields are not zeros or approval.

Actual date/version	Independent joint tree	Posterior/action error	Cost/timing correction	Objection/evidence	Revised decision	Delayed transfer

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Baseline joint/posterior/foldback	PASS:baseline joint/posterior/optimal foldback and value formulas independently checked.
Changed priors and payoff boundaries	PASS:changed priors0/1/interior, payoffs including zero endpoints.
Informative/inverted/perfect/zero signals	PASS:perfect, inverted, uninformative and impossible-signal cases.
Cost tie, high cost and delay	PASS:costs high/zero, delay, negative ceiling and exact cost tie.
Impossible signal undefined branch	PASS:zero-probability posterior UNDEFINED/action UNREACHED, weighted0.
Invalid models/cost isolation	PASS:14 invalid cases held; bad costs preserve gross arithmetic.
Broken critical references/holds	PASS:7 broken critical references held affected summary.
Restored baseline/actual work blank	PASS:24 changed sets+restore; actual learner fields blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.